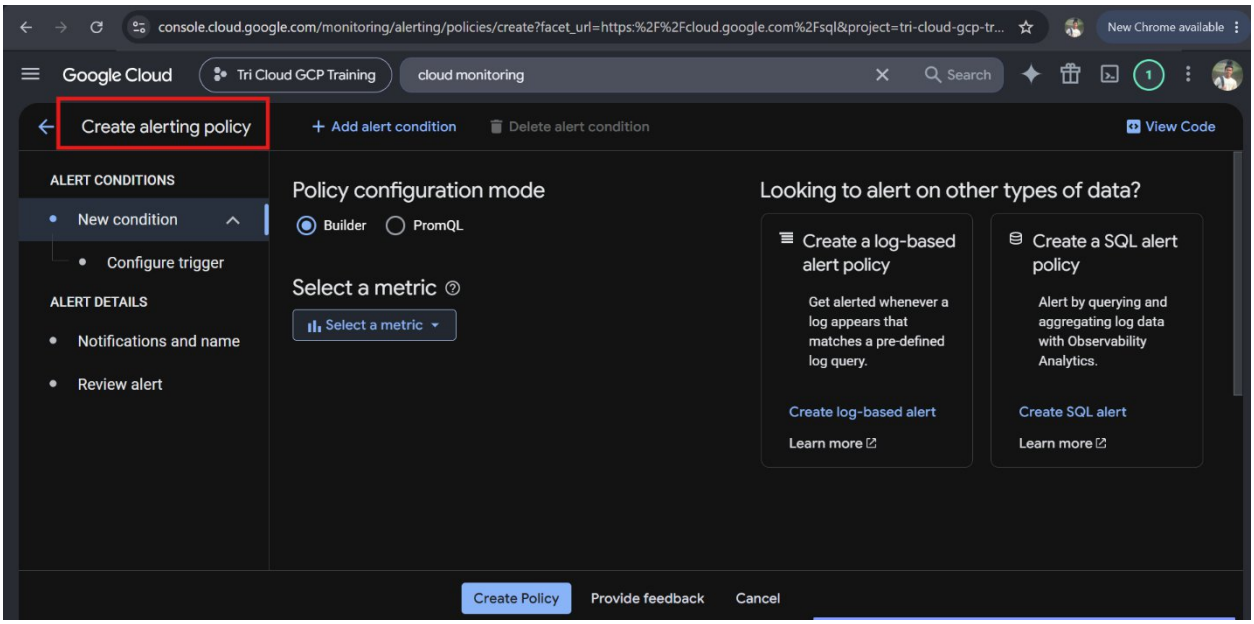


TRI CLOUD ESSENTIALS — GCP SESSION 2

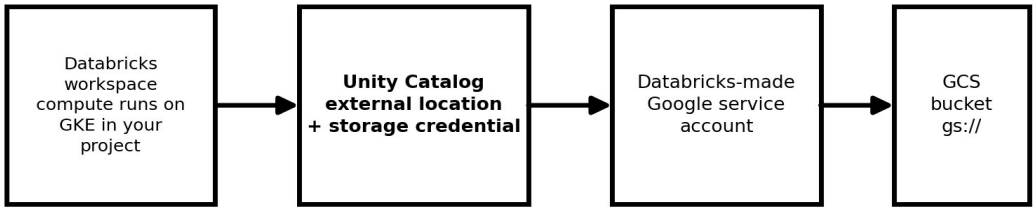
- **Topic:** GCP — Cross-Cloud Overview
- **Date:** 5th November 2026

5. Databricks Connectivity on GCP: An Overview

Databricks is also available on Google Cloud. The workspace is created in a GCP project, and its compute runs on Google Kubernetes Engine (GKE) in your own project. Like on Azure and AWS, Databricks needs a safe way to read the buckets of the data lake, and the answer is again an identity with limited roles and no downloaded key.

Way to reach Cloud Storage	Simple meaning
Unity Catalog storage credential	Databricks creates a Google service account for the credential. You give that account roles on the bucket, and then you create an external location with a gs:// path. Access is granted per group and per path. This is the recommended way.

Databricks on GCP reaching Cloud Storage



You grant that service account roles on the bucket, and no key file is needed

Diagram 7: Unity Catalog uses a Databricks-made service account that has roles on the bucket.

Step	What to do
1. Have the bucket ready	Use a Cloud Storage bucket such as tricloud-datalake-dev-001 in the project.
2. Create the credential	In Databricks open Catalog, then External data, then Credentials, and create a storage credential. Databricks shows the email of the service account it made.
3. Grant roles on the bucket	In GCP give that service account the Storage Legacy Bucket Reader role and the Storage Object Admin role on the bucket.
4. Create the external location	Create an external location with the gs:// path and the storage credential.
5. Grant and test	Give READ FILES or WRITE FILES to a group, and list the path to check it works.

6. Comparing Azure, AWS, and GCP: Five Core Competencies

In simple words. The three clouds solve the same problems with different names. If you understand an idea in one cloud, you only need to learn the new name and the small differences in the others. The five areas below are the ones used in the last three weeks.

1. Identity and Access Management. All three clouds decide who can do what on which resource, and all three prefer short-lived credentials to long-lived keys.

Point	Identity and Access Management
People and groups	Azure: Microsoft Entra ID users and groups. AWS: IAM users and groups, or IAM Identity Center. GCP: Google accounts and Google groups, from Cloud Identity or Workspace.
Identity for an app	Azure: managed identity. AWS: IAM role, which is assumed for temporary credentials. GCP: service account, which is attached or impersonated.
How permissions are given	Azure: RBAC role assignment of a principal, a role, and a scope. AWS: JSON policies attached to identities or to resources. GCP: roles bound to principals in an allow policy on a resource, inherited downward.

2. Cloud Storage. Each cloud has object storage that is the base of a data lake.

Point	Cloud Storage
Top container	Azure: a container in a storage account (ADLS Gen2). AWS: an S3 bucket. GCP: a Cloud Storage bucket.
Path form	Azure: abfss://container@account.dfs.core.windows.net/path. AWS: s3://bucket/key. GCP: gs://bucket/object.
Folders	Azure: real folders with the hierarchical namespace. AWS: prefixes only. GCP: prefixes, or real folders when the hierarchical namespace is on.
Data tiers	Azure: Hot, Cool, Cold, Archive. AWS: Standard, Infrequent Access, and Glacier classes. GCP: Standard, Nearline, Coldline, Archive.

3. Secrets Management. Each cloud stores secrets in a managed service and gives them to applications that have the right permission.

Point	Secrets Management

Service	Azure: Key Vault. AWS: Secrets Manager, with KMS for the keys. GCP: Secret Manager, with Cloud KMS for the keys.
Permission to read	Azure: the Key Vault Secrets User role. AWS: secretsmanager:GetSecretValue in a policy. GCP: the Secret Manager Secret Accessor role.
Rotation	Azure: events near expiry, and a rotation policy for keys. AWS: automatic rotation with a Lambda function. GCP: a schedule that notifies you through Pub/Sub, and your code rotates.
Keys and certificates	Azure: held in the same Key Vault. AWS: KMS for keys, Certificate Manager for certificates. GCP: Cloud KMS for keys, Certificate Manager for certificates.

4. Monitoring. Each cloud collects metrics and logs, lets you query them, and sends alerts.

Point	**Monitoring**
Metrics and alerts	Azure: Azure Monitor metrics, and alert rules with action groups. AWS: CloudWatch metrics, and alarms with SNS topics. GCP: Cloud Monitoring metrics, and alerting policies with notification channels.
Log search	Azure: Log Analytics with KQL. AWS: CloudWatch Logs with Logs Insights. GCP: Cloud Logging with the Logging query language.
Audit of actions	Azure: the Activity log. AWS: CloudTrail. GCP: Cloud Audit Logs.
Turning logs on	Azure: diagnostic settings for each resource. AWS: enabled per service, and the agent on servers. GCP: Admin Activity is always on, and Data Access logs must be enabled.

5. Databricks Connectivity. In all three, Unity Catalog uses a storage credential and an external location. Only the identity behind the credential differs.

Point	**Databricks Connectivity**
Identity behind the credential	Azure: the Access Connector, which holds a managed identity. AWS: an IAM role that Unity Catalog assumes. GCP: a Google service account made by Databricks.
Permission on storage	Azure: Storage Blob Data Contributor on the storage account. AWS: an S3 policy on the IAM role.

	GCP: Storage Legacy Bucket Reader and Storage Object Admin on the bucket.
External location path	Azure: abfss://. AWS: s3://. GCP: gs://.

At a glance. This table lines up the same idea across the three clouds.

Concept	**Azure**	**AWS**	**GCP**
Main boundary	Subscription	Account	Project
Identity for an app	Managed identity	IAM role	Service account
Permission model	RBAC role assignment	IAM policy	IAM role binding
Object storage	Storage account and container	S3 bucket	Cloud Storage bucket
Secrets	Key Vault	Secrets Manager	Secret Manager
Encryption keys	Key Vault keys	KMS	Cloud KMS
Metrics and alerts	Azure Monitor	CloudWatch	Cloud Monitoring
Log search	Log Analytics and KQL	Logs Insights	Logs Explorer
Audit trail	Activity log	CloudTrail	Cloud Audit Logs
Databricks storage link	Access Connector	IAM role credential	Service account credential

- **Same ideas, new names.** Least privilege, short-lived credentials, secrets kept outside code, and alerts on failures hold in every cloud.
- **Main differences to remember.** AWS puts policies on identities, GCP binds roles on resources, and Azure assigns roles at a scope. Azure Key Vault holds secrets, keys, and certificates together, and the other two split them.
- **Real folders.** ADLS Gen2 and Cloud Storage with the hierarchical namespace on have real folders. S3 is flat, and so is Cloud Storage without that setting.